



Innovative Tech Solutions for
Health

Smart Band for Comprehensive Health Monitoring

**Microprocessor Systems Course: Exploring the
integration of IoT in health monitoring devices**

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Project Overview and Features

This agenda outlines the key features and goals of the smart band project for health monitoring.



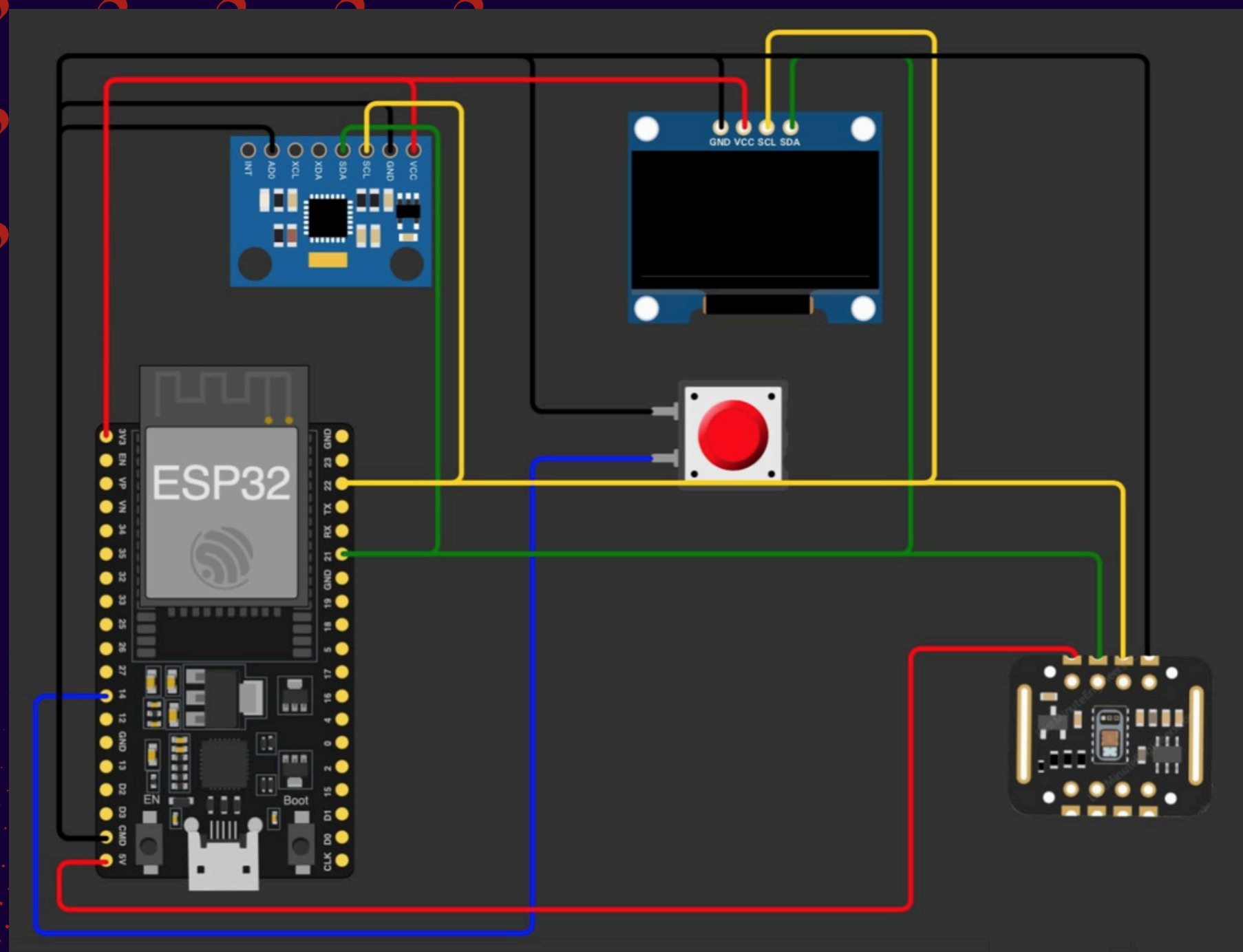
Overview of health metrics tracked: HR, SpO₂, Temp, and Steps.

Technology utilized: ESP32 with MAX30105 and MPU6050 sensors for data collection.

Data transmission methods: Wi-Fi and MQTT to a Flask server for processing.

User interface: Visualizing health data on a web dashboard for easy access.

Prototype objectives: Achieving a low-cost solution for comprehensive health monitoring.



Understanding the System Architecture for the Smart Band Health Monitoring

System Integration Overview

This architecture illustrates how the smart band collects and transmits health data efficiently.

By utilizing the ESP32 and various sensors, data flows seamlessly through MQTT to the Flask server, ensuring real-time visualization on the web dashboard for users to monitor their health metrics effectively.

 **HIVEMQ**

Websockets Client Showcase

 **HIVEMQ** CLOUD

Need a fully managed MQTT broker?
Get your own Cloud broker and connect up to 100 devices for free.

Get your free account

Connection

connected

Host
broker.hivemq.com

Port
8884

ClientID
clientId-qT8IEqKH4D

Disconnect

Username

Password

Keep Alive
1000

SSL
x

Clean Session
x

Last-Will Topic

Last-Will QoS
0

Last-Will Retain

Last-Will Message

Publish

Topic
testtopic/1

QoS
0

Retain

Publish

Message

Subscriptions

Add New Topic Subscription

MQTT Protocol for IoT Data Transmission

Protocol Used: MQTT (Message Queuing Telemetry Transport)

Broker: HiveMQ (cloud-based)

Why MQTT?

Lightweight & efficient for IoT

Low bandwidth usage

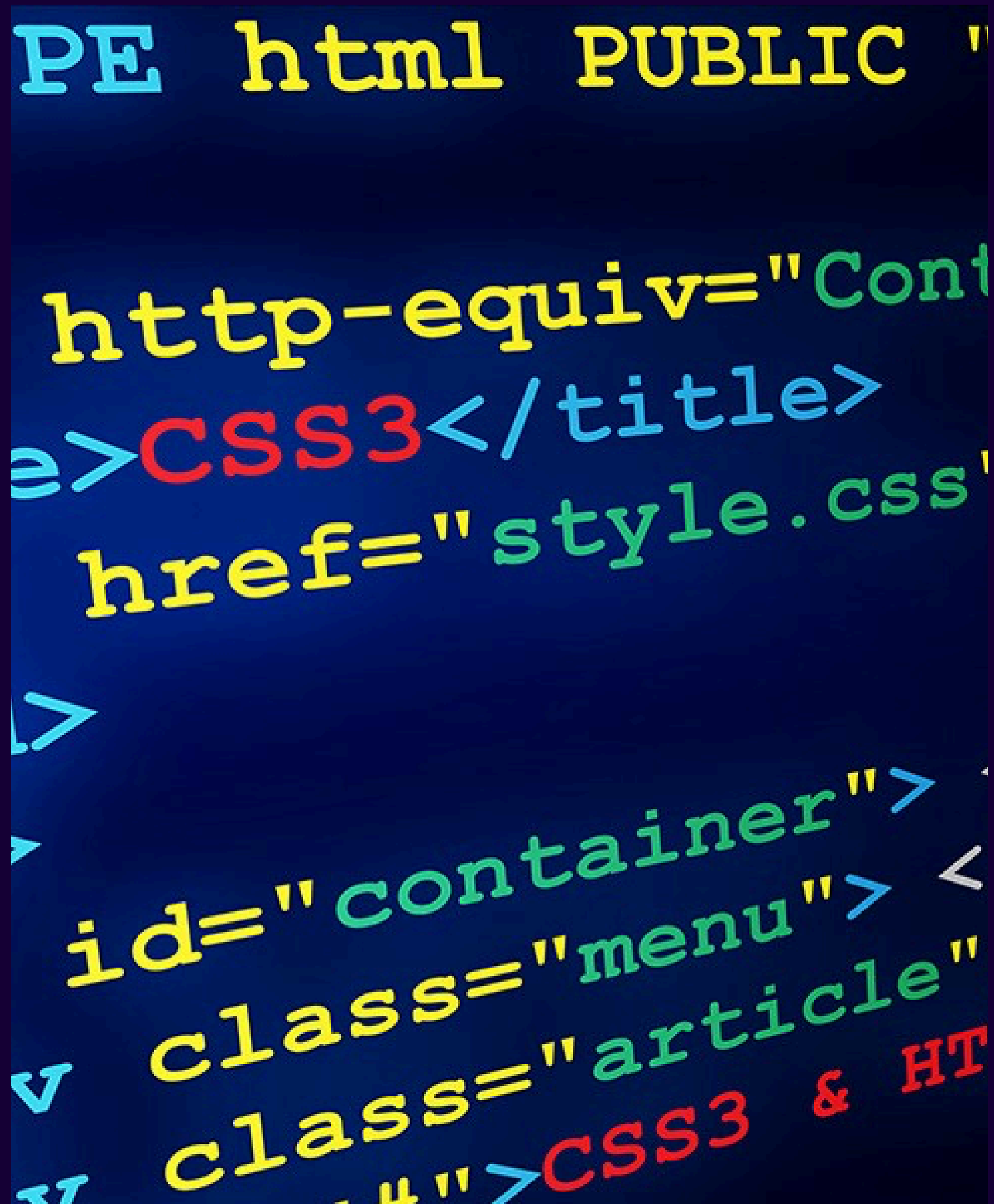
Reliable delivery using publish/subscribe model

Functionality:

ESP32 publishes health data in JSON to topic esp32/bruh/data

Flask backend subscribes and stores the data in user-specific CSV files

MQTT ACK used for confirming signup



HTML Frontend for Health Data Visualization

- **Pages Developed:**

dashboard.html: Displays real-time sensor data in table form

graphs.html: Visualizes HR, SpO₂, Temp, and Steps using Chart.js

- **Technologies Used:**

HTML, CSS (Bootstrap), JavaScript
Chart.js for interactive graphs

- **User Features:**

Easy-to-use interface

Auto-refreshing data display

Fitness metrics in fitness.html

Login

Username

Password

Login

[Don't have an account? Sign Up](#)

Sign Up

Watch ID

Username

Password

Age

Weight (kg)

Height (cm)

Gender

Sign Up

[Already have an account? Login](#)

Secure User Authentication (Login & Signup)

Flask Backend Features:

signup.html & login.html forms

User feedback with flash messages

MQTT ACK used to validate device ownership during signup

Functionality:

Users register with watch ID

Upon login, users view their personalized dashboard

CSV file created for each user to store data

Frontend Dashboard: Engaging Visual Representation of Health Data for User Interaction

Visual Data Insights

The frontend dashboard displays health metrics in a user-friendly format for easy monitoring.

With real-time updates and visually appealing graphs, users can track their health data effectively and make informed decisions about their well-being.

Welcome, ID: 1001

View Graphs

View Fitness Dashboard

Timestamp	Steps	HeartRate	Temperature	Oxygen
2025-05-30 12:08:50	10	46	28.60353	67.95866
2025-05-30 12:08:55	10	46	28.61823	67.72152
2025-05-30 12:08:55	10	46	28.61823	67.72152
2025-05-30 12:09:00	10	46	28.58882	66.4557
2025-05-30 12:09:00	10	46	28.58882	66.4557
2025-05-30 12:09:05	10	46	28.55941	68.61219
2025-05-30 12:09:05	10	46	28.55941	68.61219
2025-05-30 12:09:10	10	46	28.52118	66.28644
2025-05-30 12:09:10	10	46	28.52118	66.28644
2025-05-30 12:09:15	10	46	28.45941	66.66666
2025-05-30 12:09:15	10	46	28.45941	66.66666
2025-05-30 12:09:20	10	46	28.44176	67.09347

Fitness Dashboard (fitness.html)

Fitness Dashboard

Workout Intensity
Low

Workout Duration (min)
30

Workout Type
General

Sleep Hours
7

Water Intake (L)
2

Body-Fat (%)
15

Mode
☒ Before Workout ☐ After Workout

Calculate

BMI: 22.9

Calories Burned: 70 kcal

Sleep: 7.0 hrs

Water Intake: 2.0 L

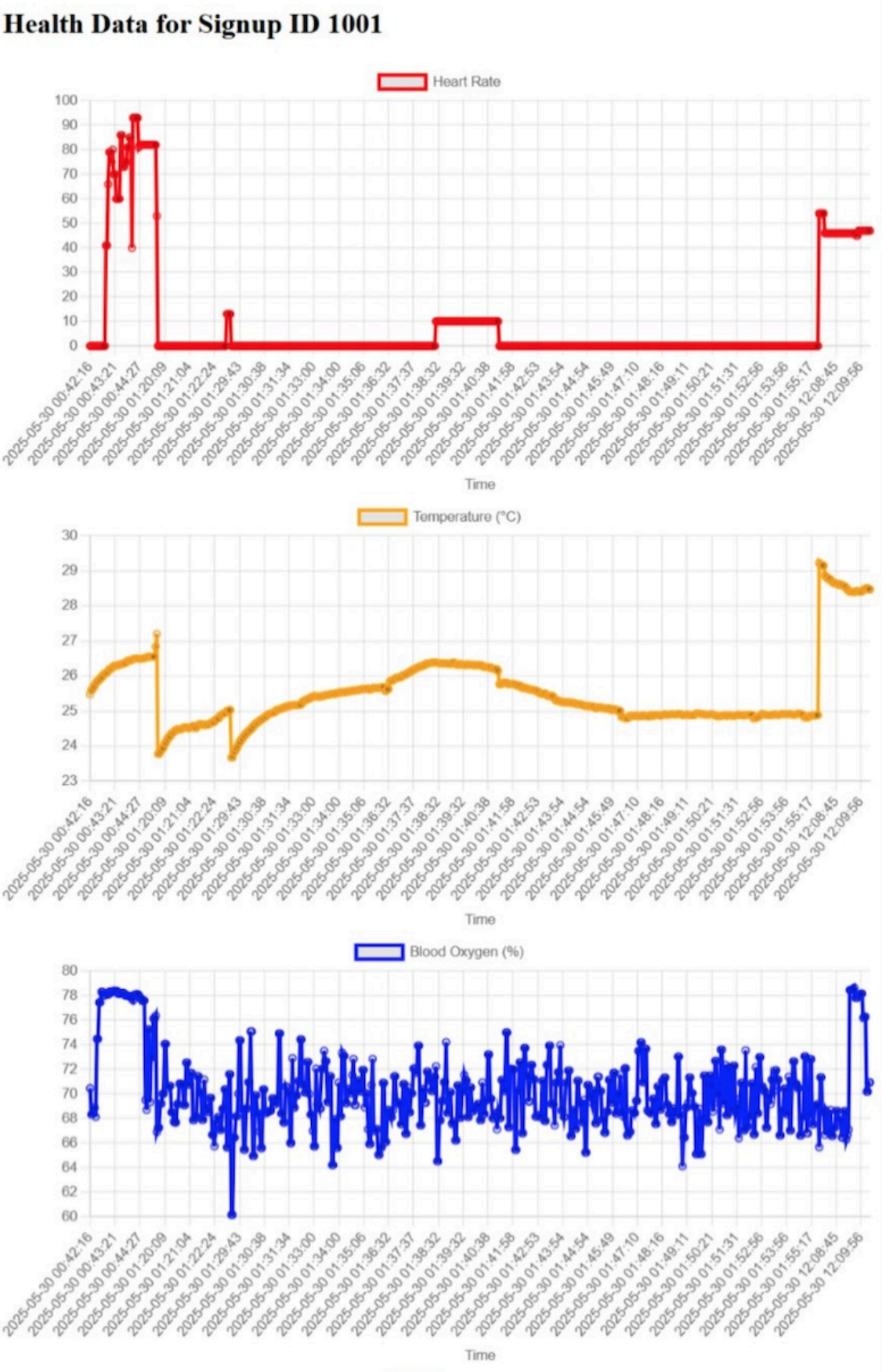
Body-Fat: 15.0 %

Mode: Before Workout



Interactive Health & Fitness Calculator

- Page Features:
 - Developed using HTML, CSS, and JavaScript
 - Fully responsive card-based layout, mobile-friendly
- User Inputs:
 - Workout type & duration
 - Intensity level & Sleep hours
 - Water intake
 - Body fat percentage
- Outputs (On Button Click):
 - BMI (Body Mass Index)
 - Calories burned
 - Health score indicators



Experimental Results Analysis

📱 Real-Time Health Data Monitoring

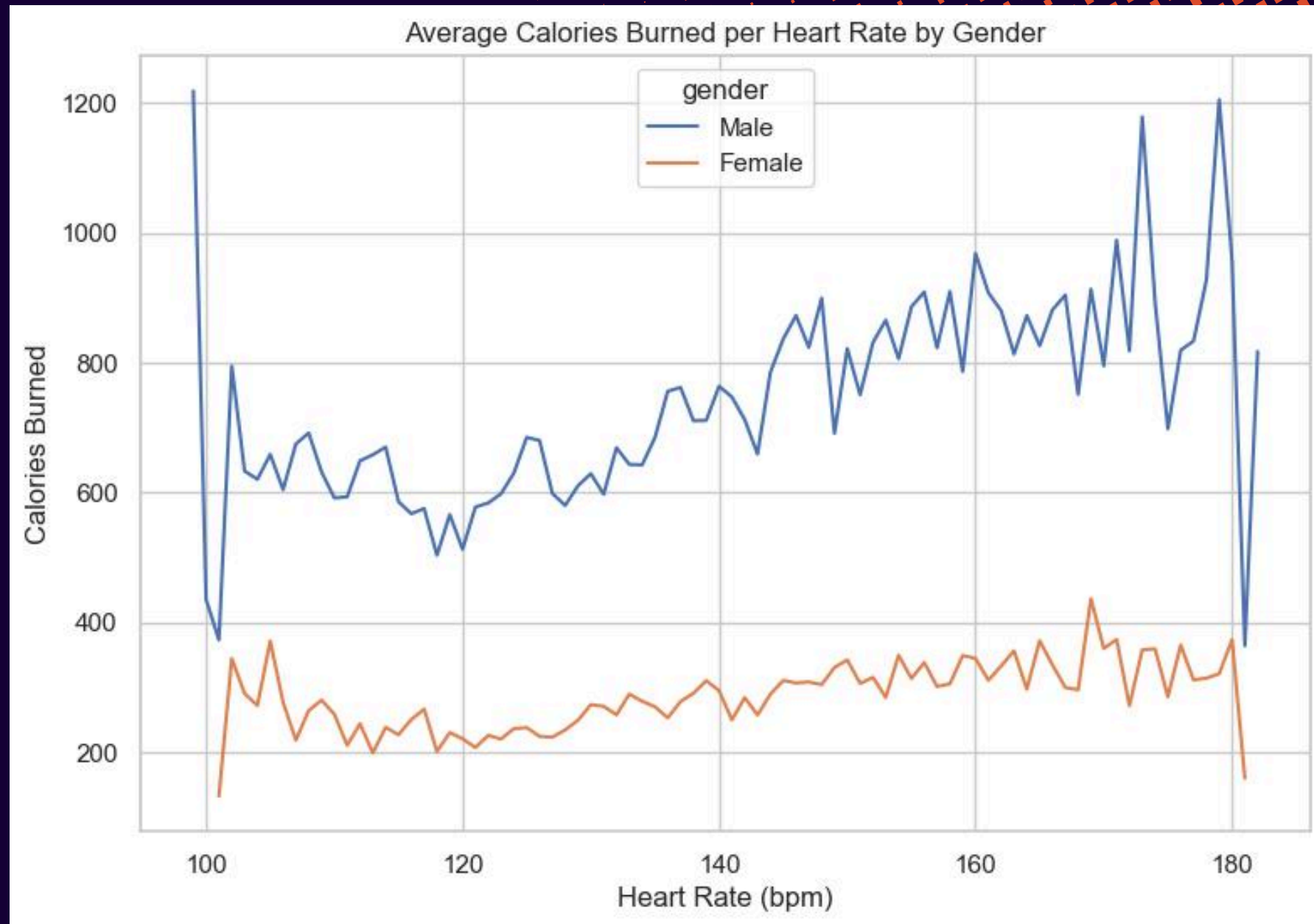
The smart band successfully collected and transmitted real-time physiological data. All sensor readings were streamed live via MQTT to the backend server. The web dashboard auto-updates and visualizes data dynamically using Chart.js.

📊 Key Observations:

Heart Rate (Red): Detects changes in activity levels over time.

Temperature (Orange): Tracks fluctuations influenced by skin and environment.

SpO₂ (Blue): Maintains stable readings, with expected minor variations



Graphical Data Insights

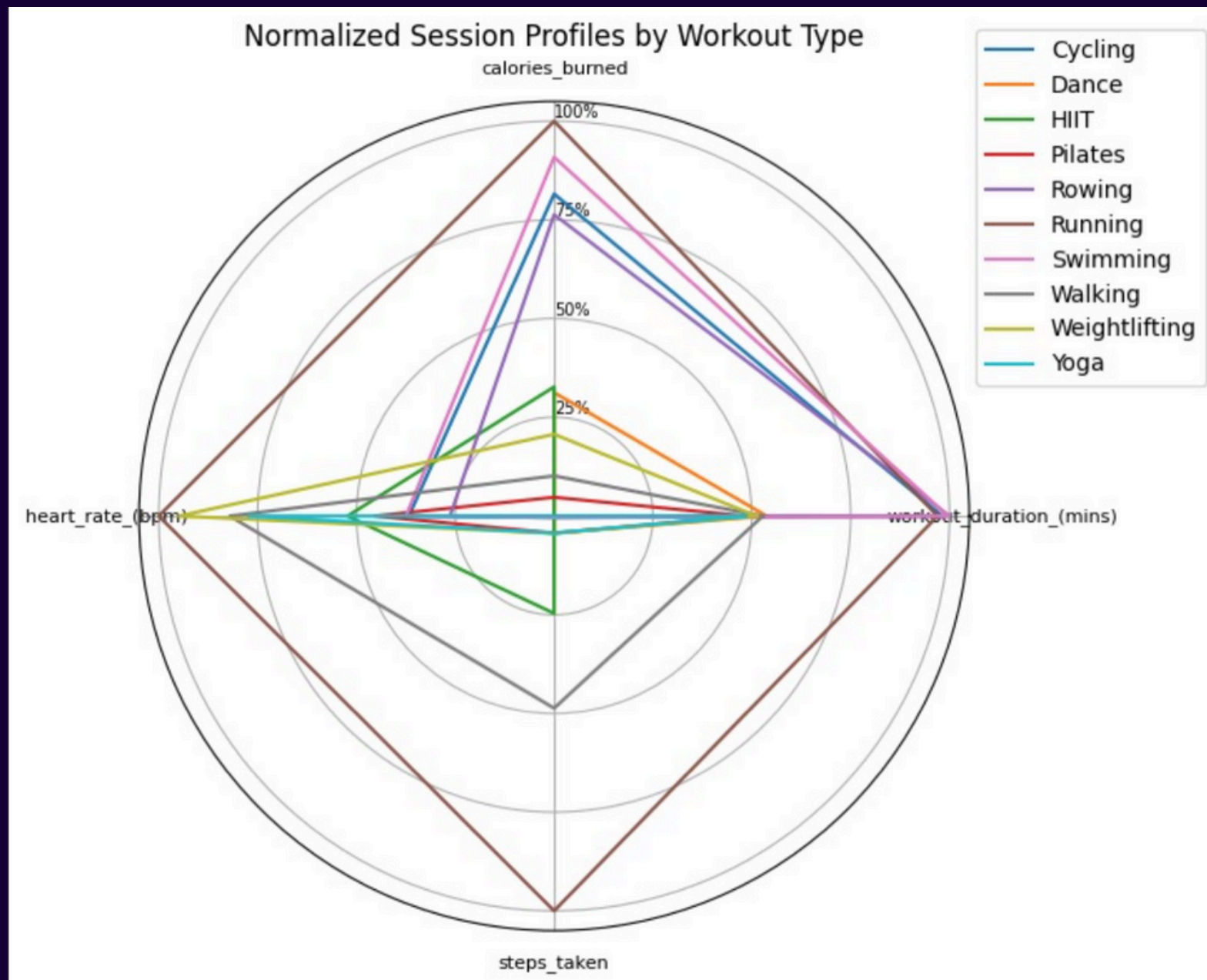
This section will explore the visual representation of Average Calories Burned per Heart Rate by Gender

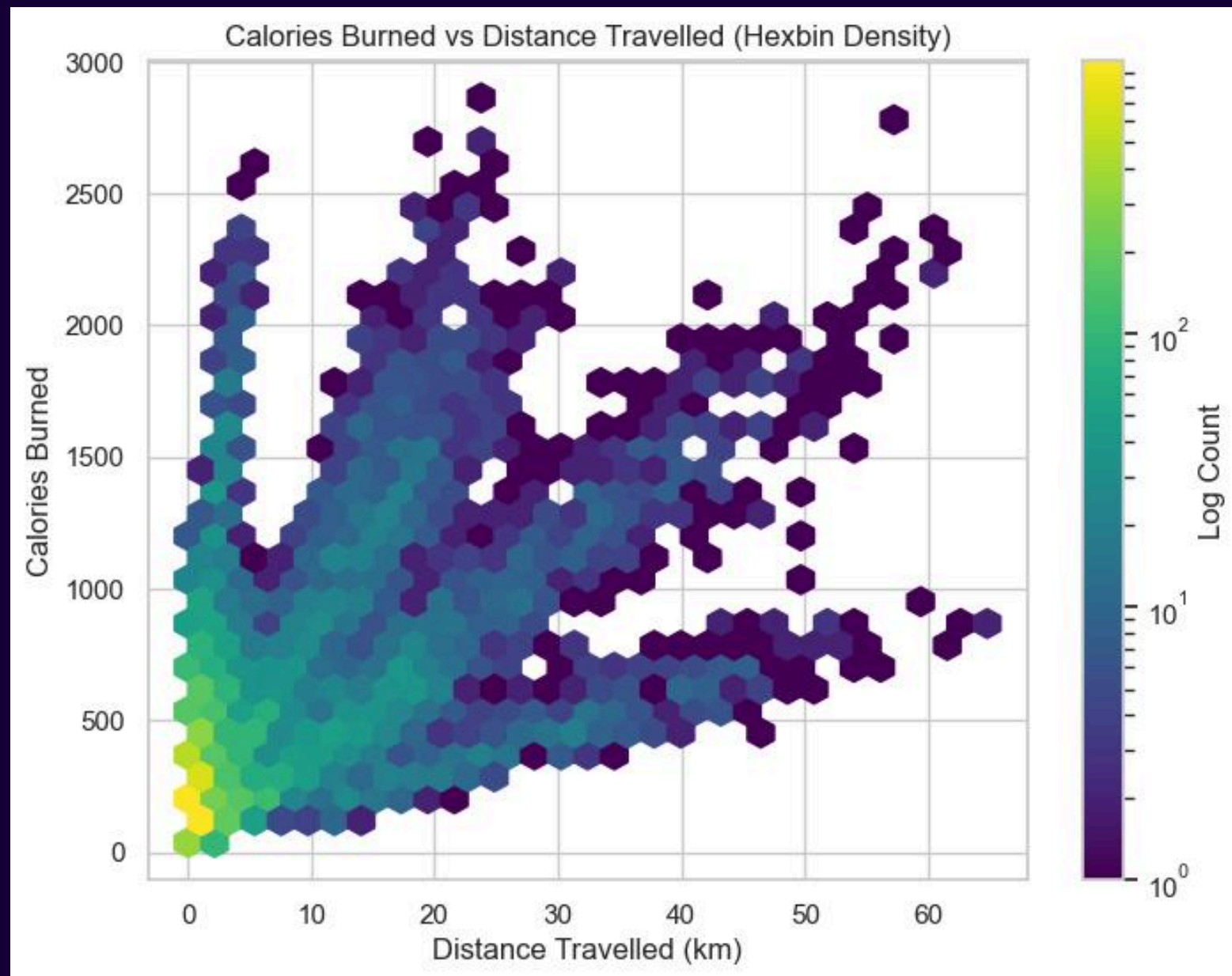
Insights: Males burn more calories per Heart Rate

Graphical Data Insights

This section will explore the visual representation of Normalized Session Profiles by Workout Type

Insights: Running stands out with the highest normalized values for calories burned, steps taken, and workout duration, indicating its overall intensity. In contrast, HIIT triggers the highest heart rate but with lower steps and duration, reflecting its short, high-intensity nature.





Graphical Data Insights

This section will explore the visual representation of Calories Burned vs Distance Travelled (Hexbin Density)

Insights: The hexbin plot shows a clear positive relationship between distance travelled and calories burned, with most data concentrated between 0–20 km and 0–1000 calories. Higher distances generally lead to more calories burned, though with increasing variability at longer distances.

Challenges & Limitations Overview

In this section, we will explore the key challenges encountered during our project development process.

Sensor placement: ensuring accurate data collection in various scenarios.

Battery life: balancing performance with long-lasting usage for wearability.

Security concerns: safeguarding user data and maintaining privacy in health monitoring.

CSV scalability: managing large datasets efficiently for analysis and storage.

User experience: addressing usability issues in the interface and functionality of the system.



Future Work and Enhancements

This section outlines potential areas for improvement and growth in our IoT-enabled health monitoring system.

Exploring Bluetooth Low Energy (BLE) for enhanced connectivity options.

Implementing Over-The-Air (OTA) updates for seamless software upgrades.

Enhancing security measures to protect user data and privacy.

Developing a mobile app to increase user engagement and accessibility.

Incorporating machine learning for predictive health analytics and insights.





Thank You